

Report Assignment 5

Group 2

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1 Introduction

Counterparty credit risk is a central issue in the valuation and risk management of OTC derivatives. Unlike traditional credit exposures, the amount at risk is not fixed in advance: it depends on the future mark-to-market of the contract, which evolves with market conditions.

In this report we consider the position of Polimi Bank in a 10-year vanilla payer interest rate swap. The objective is to estimate the future exposure generated by this trade from the bank's perspective. To do so, we simulate the evolution of the interest-rate curve under the one-factor Hull–White model. The model is designed to fit the initial market discount curve exactly, while introducing a stochastic mean-reverting component for the future evolution of short rates.

The simulated future discount curves are then used to revalue the swap across Monte Carlo scenarios and future simulation dates. This produces a distribution of future mark-to-market values, from which the relevant counterparty-risk quantities can be computed. The analysis therefore links interest-rate modelling, swap valuation, and exposure measurement in a single simulation framework.

2 Discount factor curve simulation

In the Hull-White model, the short rate is decomposed as

$$r_t = \varphi(t) + x_t,$$

where $\varphi(t)$ is deterministic and the stochastic component follows the Ornstein–Uhlenbeck dynamics

$$dx_t = -ax_t dt + \sigma dW_t, \quad a > 0, \quad \sigma > 0.$$

The deterministic function $\varphi(t)$ is chosen so that the model exactly fits the initial market discount curve observed at t_0 .

We define the auxiliary process

$$x'_t = e^{at} x_t.$$

By applying Itô's formula to x'_t , we obtain the solution of the Ornstein–Uhlenbeck process between two dates s and t :

$$x_t = x_s e^{-a(t-s)} + \sigma \int_s^t e^{-a(t-u)} dW_u.$$

Hence x_t is Gaussian and mean-reverting. In the Monte Carlo simulation, this is the state variable driving the whole discount curve.

The stochastic discount factor between s and τ is

$$D(s, \tau) = \exp \left(- \int_s^\tau r_u du \right).$$

Using the solution of the Ornstein–Uhlenbeck process and Fubini–Tonelli's theorem, the integral of the short rate can be written in terms of x_s and of a stochastic integral with deterministic volatility. Defining

$$\sigma(u, \tau) = \sigma \frac{1 - e^{-a(\tau-u)}}{a},$$

we obtain

$$D(s, \tau) = \exp \left\{ - \int_s^\tau \varphi(u) du - x_s \frac{\sigma(s, \tau)}{\sigma} - \int_s^\tau \sigma(u, \tau) dW_u \right\}.$$

As one can notice, the stochastic integral

$$\int_s^\tau \sigma(u, \tau) dW_u$$

is Gaussian, since the integrand is deterministic. Its conditional mean and variance are

$$\mathbb{E}_s \left[\int_s^\tau \sigma(u, \tau) dW_u \right] = 0, \quad \text{Var}_s \left[\int_s^\tau \sigma(u, \tau) dW_u \right] = \int_s^\tau \sigma^2(u, \tau) du.$$

Therefore, by taking the conditional expectation of the stochastic discount factor,

$$B(s, \tau) = \mathbb{E}_s[D(s, \tau)],$$

we obtain

$$B(s, \tau) = \exp \left\{ - \int_s^\tau \varphi(u) du - x_s \frac{\sigma(s, \tau)}{\sigma} + \frac{1}{2} \int_s^\tau \sigma^2(u, \tau) du \right\}.$$

Starting from the forward discount factor

$$B(s, \tau, \tau + \theta) = \frac{B(s, \tau + \theta)}{B(s, \tau)}.$$

and using the expression obtained for $B(s, T)$, we can write

$$B(s, \tau, \tau + \theta) = \exp \left\{ - \int_\tau^{\tau + \theta} \varphi(u) du - x_s \left(\frac{\sigma(s, \tau + \theta) - \sigma(s, \tau)}{\sigma} \right) + \frac{1}{2} \int_s^{\tau + \theta} \sigma^2(u, \tau + \theta) du - \frac{1}{2} \int_s^\tau \sigma^2(u, \tau) du \right\}.$$

At the initial date t_0 , since $x_{t_0} = 0$, the same quantity is

$$B(t_0, \tau, \tau + \theta) = \exp \left\{ - \int_\tau^{\tau + \theta} \varphi(u) du + \frac{1}{2} \int_{t_0}^{\tau + \theta} \sigma^2(u, \tau + \theta) du - \frac{1}{2} \int_{t_0}^\tau \sigma^2(u, \tau) du \right\}.$$

Taking the ratio between the two forward discount factors gives

$$\frac{B(s, \tau, \tau + \theta)}{B(t_0, \tau, \tau + \theta)} = \exp \left\{ -x_s \left(\frac{\sigma(s, \tau + \theta) - \sigma(s, \tau)}{\sigma} \right) - \frac{1}{2} \int_{t_0}^s [\sigma^2(u, \tau + \theta) - \sigma^2(u, \tau)] du \right\}.$$

The key point is that the deterministic drift-adjustment term φ cancels out in the ratio:

$$- \int_\tau^{\tau + \theta} \varphi(u) du + \int_\tau^{\tau + \theta} \varphi(u) du = 0.$$

Therefore the forward discount factor can be reconstructed using only the initial market curve and the simulated state variable x_s :

$$B(s, \tau, \tau + \theta) = B(t_0, \tau, \tau + \theta) \exp \left\{ -x_s \left(\frac{\sigma(s, \tau + \theta) - \sigma(s, \tau)}{\sigma} \right) - \frac{1}{2} \int_{t_0}^s [\sigma^2(u, \tau + \theta) - \sigma^2(u, \tau)] du \right\}.$$

We now use this result to obtain the discount factor $B(s, \tau)$. Since

$$B(s, s, \tau) = \frac{B(s, \tau)}{B(s, s)} = B(s, \tau),$$

and

$$B(t_0, s, \tau) = \frac{B(t_0, \tau)}{B(t_0, s)},$$

we obtain

$$B(s, \tau) = \frac{B(t_0, \tau)}{B(t_0, s)} \exp \left\{ -x_s \left(\frac{\sigma(s, \tau) - \sigma(s, s)}{\sigma} \right) - \frac{1}{2} \int_{t_0}^s [\sigma^2(u, \tau) - \sigma^2(u, s)] du \right\}.$$

Now, since

$$\sigma(s, s) = 0,$$

and

$$\frac{\sigma(s, \tau) - \sigma(s, s)}{\sigma} = \frac{1 - e^{-a(\tau-s)}}{a},$$

we finally get

$$B(s, \tau) = \frac{B(t_0, \tau)}{B(t_0, s)} \exp \left\{ -\frac{1}{2} \int_{t_0}^s [\sigma^2(u, \tau) - \sigma^2(u, s)] du \right\} \exp \left\{ -x_s \frac{1 - e^{-a(\tau-s)}}{a} \right\}.$$

Therefore the discount factor curve at each simulation date s can be written in affine form as

$$B(s, \tau) = A(s, \tau) e^{-x_s C(s, \tau)},$$

where

$$C(s, \tau) = \frac{1 - e^{-a(\tau-s)}}{a},$$

and

$$A(s, \tau) = \frac{B(t_0, \tau)}{B(t_0, s)} \exp \left\{ -\frac{1}{2} \int_{t_0}^s [\sigma^2(u, \tau) - \sigma^2(u, s)] du \right\}.$$

The integral above admits a closed-form solution. Following Brigo and Mercurio (*Interest Rate Models — Theory and Practice*, 2nd ed., 2006, eq. 3.39, p. 75), the function $A(s, \tau)$ can be written explicitly as:

$$A(s, \tau) = \frac{B(t_0, \tau)}{B(t_0, s)} \exp \left\{ -\frac{1}{2} C(s, \tau)^2 \cdot \frac{\sigma^2}{2a} (1 - e^{-2a(s-t_0)}) \right\},$$

Hence, for each Monte Carlo scenario and for each simulation date s , once the state variable x_s has been simulated, the whole discount curve is reconstructed as

$$\boxed{B(s, \tau) = A(s, \tau) e^{-x_s C(s, \tau)}}.$$

3 Swap mark-to-market simulation

Once the future discount factor curves have been reconstructed, we compute the mark-to-market of the 10-year payer interest rate swap at each simulation date and for each Monte Carlo scenario.

The swap is assumed to be entered at par at the initial date. Therefore, the fixed rate K is computed from the initial discount curve as the par swap rate:

$$K = \frac{1 - B(t_0, T_N)}{\sum_{i=1}^N \delta_i B(t_0, T_i)},$$

where $T_1, \dots, T_N$ are the fixed-leg payment dates and δ_i are the corresponding year fractions.

At a future simulation date s , only the remaining cash flows are relevant for the swap valuation. For each scenario, the simulated discount curve

$$\{B(s, T_i)\}_{T_i > s}$$

is used to compute the present value of the fixed and floating legs.

The present value of the fixed leg is

$$PV_{\text{fix}}(s) = K \sum_{T_i > s} \delta_i B(s, T_i).$$

Under the single-curve framework, the value of the remaining floating leg after a reset date is

$$PV_{\text{float}}(s) = 1 - B(s, T_N),$$

where T_N is the final maturity of the swap. This expression is applied only while the swap is alive; after maturity the swap value is set equal to zero.

Since Polimi Bank enters a payer swap, it pays the fixed leg and receives the floating leg. Therefore, the mark-to-market from the bank's perspective is

$$MtM(s) = N (PV_{\text{float}}(s) - PV_{\text{fix}}(s)),$$

that is

$$MtM(s) = N \left[1 - B(s, T_N) - K \sum_{T_i > s} \delta_i B(s, T_i) \right],$$

where N is the swap notional.

In the Monte Carlo implementation, this calculation is performed for every simulation date s and every scenario. The matrix of simulated discount factors has one row per scenario and one column per future date. Therefore, the fixed leg is obtained by summing the discounted fixed cash flows across columns, while the floating leg is obtained from the simulated discount factor corresponding to the swap maturity.

The resulting simulated mark-to-market matrix is the input for the exposure calculation. Positive values represent scenarios in which the swap is an asset for Polimi Bank and therefore generate counterparty exposure; negative values represent scenarios in which the swap is a liability and do not contribute to positive counterparty exposure.

4 EPE & PFE calculation

4.1 EPE

The **Expected Positive Exposure** at a future date s is defined as

$$\text{EPE}(s) = \mathbb{E}_0[\text{Exposure}_s^+] = \mathbb{E}_0[\max(\text{MtM}_s - \text{Collateral}_s, 0)], \quad (1)$$

The EPE represents the average amount that Polimi Bank can lose in the event that the counterparty defaults at time s .

For each simulation date s_i ($i = 1, \dots, 40$) the EPE is approximated by the cross-sectional sample mean of the positive exposures across the N_{sim} scenarios:

$$\widehat{\text{EPE}}(s_i) = \frac{1}{N_{\text{sim}}} \sum_{k=1}^{N_{\text{sim}}} \max(\text{MtM}_{s_i}^{(k)} - \text{Collateral}_{s_i}^{(k)}, 0). \quad (2)$$

4.2 PFE

The **Potential Future Exposure** at confidence level α and future date s is the α -quantile of the positive-exposure distribution:

$$\text{PFE}_\alpha(s) = Q_\alpha[\text{Exposure}_s^+] = Q_\alpha[\max(\text{MtM}_s - \text{Collateral}_s, 0)], \quad (3)$$

where $Q_\alpha[\cdot]$ denotes the α -quantile of the cross-sectional distribution across scenarios. In this assignment $\alpha = 95\%$. The PFE is the CCR analogue of Value-at-Risk for market risk: it represents a *severe but plausible* exposure scenario rather than an average.

For each simulation date s_i the 95% PFE is computed as the empirical 95th percentile of the positive-exposure distribution:

$$\widehat{\text{PFE}}_{0.95}(s_i) = Q_{0.95}[\{\max(\text{MtM}_{s_i}^{(k)} - \text{Collateral}_{s_i}^{(k)}, 0)\}_{k=1}^{N_{\text{sim}}}], \quad (4)$$

4.3 Peak-PFE

The **Peak-PFE** is defined as the supremum of the PFE profile over the entire simulation horizon $[0, T]$:

$$\text{Peak-PFE}_\alpha = \sup_{s \in [0, T]} \text{PFE}_\alpha(s), \quad (5)$$

together with the date s^* at which this maximum is attained:

$$s^* = \arg \max_{s_i} \text{PFE}_\alpha(s_i). \quad (6)$$

The Peak-PFE is a single scalar that summarises the worst-case exposure level at a given confidence level over the entire life of the transaction. Because credit limits are typically expressed as scalar quantities rather than time-dependent profiles, financial institutions compare the Peak-PFE to the approved counterparty limit to ensure that exposures never exceed the defined risk appetite at any point in time. As such, the Peak-PFE is a conservative metric primarily used for credit limit setting and high-level risk reporting.

4.4 Results: Base Case

Figure 1a and Figure 1b show the EPE and PFE profiles for the base case, i.e. the standalone 10-year payer IRS without any credit mitigation.

Both profiles exhibit a distinctive sawtooth pattern, which is fundamentally linked to the discrete nature of the annual coupon payments. As the simulation moves closer to a payment date, the exposure rises as the value of the upcoming cash flow accumulates within the contract (accrual effect). Once the coupon is settled, the potential exposure drops sharply, as that cash flow is no longer a future obligation.

The PFE is significantly higher than the EPE at every simulation date, as it targets the 95th percentile of the positive-exposure distribution rather than its mean. The Peak-PFE of approximately €14.9M materialises in early 2012, and represents the worst-case counterparty exposure at the 95% confidence level over the entire life of the trade.

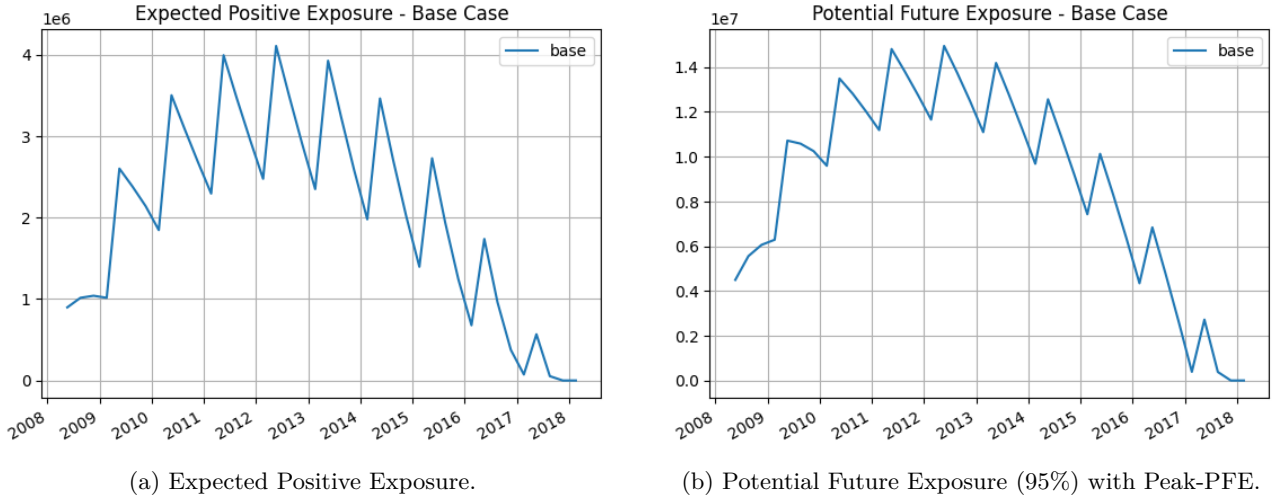


Figure 1: Base Case – EPE and PFE profiles.

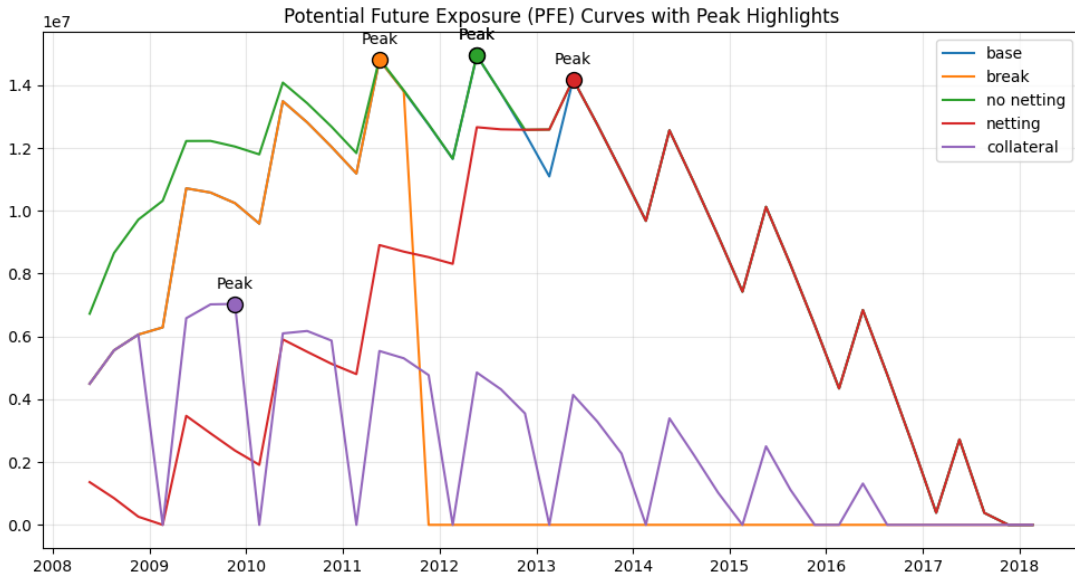


Figure 2: Peak - PFE profiles for all the cases under study

5 Netting Agreements

5.1 Point a: Mandatory Break Clause

A mandatory break clause is a contractual provision that requires the termination of a derivative transaction at a pre-specified date. Upon activation, the trade is settled at its prevailing mark-to-market (MtM) value, and all future obligations cease to exist.

The primary ratio of this clause is the significant reduction of counterparty credit risk and the achievement of regulatory capital relief. From a regulatory perspective, Basel rules generally grant capital relief only to mandatory breaks because they can be modeled deterministically, effectively shortening the legal maturity of the exposure.

In our numerical implementation, the effect of the 3.5y mandatory break clause has been modeled by setting the swap's MtM value to zero for all simulation dates following the break date. This reflects the fact that no exposure can exist once the contract has been legally terminated and settled.

5.2 Point b: Netting Agreement

The primary ratio for establishing a close-out netting agreement, typically governed by the ISDA Master Agreement, is the drastic reduction of counterparty credit risk. Netting provides a legal framework that allows all assets and liabilities with a counterparty to be collapsed into a single net amount in the event of default.

While a standard **hedging** strategy focuses on offsetting market risk to keep the portfolio value stable, it does not inherently protect against credit risk without a legal netting agreement. In a default scenario without netting:

- The surviving party must settle the gross amount owed on positions where it has a liability.
- For positions where it is owed money (assets), it must file a bankruptcy claim, often leading to minimal recovery after long legal proceedings.

This asymmetry creates a "gross exposure" that is far higher than the economic reality of the hedge. Conversely, with netting, the cash flows are offset, and only the single net balance is exchanged, effectively turning an economic hedge into a legally enforceable credit hedge.

In our exercise, we modeled this by introducing a 5-year receiver swap as an offsetting position to the original 10-year payer swap. At each of the 40 simulation dates, we calculated the MtM of this new contract and added it to the MtM of the naked 10-year payer position. The receiver swap matures after 5 years, after which it no longer contributes to the portfolio value. Therefore, its mark-to-market value is defined as:

$$MtM_{5yReceiver}(t) = 0 \quad \forall t > 5y \quad (7)$$

Consequently, for all simulation dates beyond the 5-year horizon, the netting set exposure effectively reverts to the exposure of the standalone 10-year payer swap. The exposure for this netting set was then computed as:

$$Exposure_{Netting}(t) = \max(MtM_{10yPayer}(t) + MtM_{5yReceiver}(t), 0) \quad (8)$$

This combined position represents a significantly lower risk profile compared to the standalone trades. By allowing the negative MtM of one swap to offset the positive MtM of the other, the netting agreement ensures that the bank's actual loss in a default scenario is confined to the net value.

5.3 Point c: Credit Support Annex (CSA) and Collateralization

A Credit Support Annex (CSA) is a legal document that regulates the exchange of collateral (cash or securities) to mitigate counterparty credit risk. By requiring the party with a negative Mark-to-Market (MtM) to post assets to the other, the unsecured exposure is drastically reduced, effectively converting credit risk into liquidity and operational risk.

The primary parameters governing this process include:

- **Threshold (THR):** The amount of exposure that remains unsecured; collateral is only posted for the portion of MtM exceeding this level.
- **Minimum Transfer Amount (MTA):** The smallest amount of collateral that can be transferred, designed to avoid the operational inefficiency of frequent, small margin calls.
- **Cap:** The maximum limit on the total collateral that a counterparty can be required to post.

In this assignment, the objective was to design a CSA where Polimi Bank (Party B) is never required to post collateral, while the counterparty (Party C) fully covers its exposure. To achieve this, we configured the following parameters:

- **Polimi Bank (B):** $THR_B = \infty$, $MTA_B = \infty$, and $Cap_B = 0$. These settings ensure that no matter how negative the MtM becomes, the threshold and transfer requirements are never met, and the cap prevents any posting.
- **Counterparty (C):** $THR_C = 0$, $MTA_C = 0$, and $Cap_C = \infty$. This ensures that every positive MtM value is immediately subject to a margin call for the full amount, without any upper limit.

The collateral update logic is implemented annually as follows:

1. **Target Calculation:** The target Variation Margin (VM^*) is calculated based on the current MtM and the respective thresholds:

$$VM^* = \begin{cases} \max(MtM - THR_C, 0) & MtM \geq 0 \\ \min(MtM + THR_B, 0) & MtM < 0 \end{cases} \quad (9)$$

2. **Margin Call:** A transfer (ΔVM) only occurs if the difference between VM^* and the previously posted collateral (VM_{t-1}) exceeds the relevant MTA:

$$\Delta VM = (VM^* - VM_{t-1}) \cdot \mathbb{I}_{|\Delta VM| \geq MTA} \quad (10)$$

3. **Remuneration:** The collateral balance is not considered static between observation dates. Specifically, the variation margin already posted at the previous step, VM_{t-1} , evolves over time at the risk-free rate (using the shortest available zero rate as a proxy) during the interval $[t-1, t]$. This ensures that the time-value of the collateral is correctly accounted for, so that the margin call ΔVM is calculated as the difference between the target VM^* and the accrued value of the previously held collateral.

5.4 Results & Comments

In this section, we analyze the EPE and PFE profiles obtained from the Monte Carlo simulation across the three different risk mitigation scenarios.

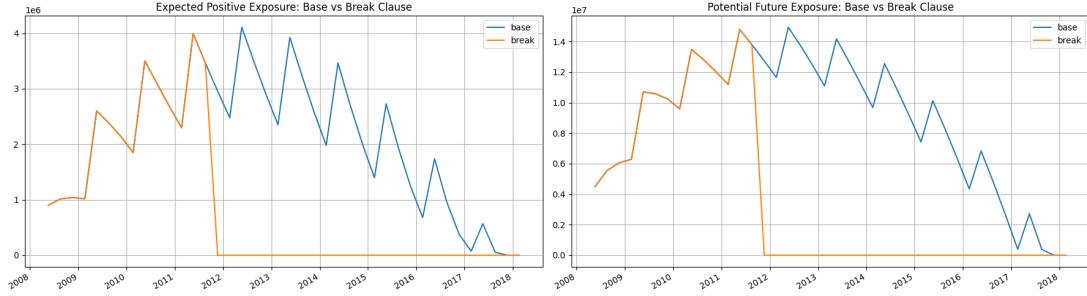


Figure 3: EPE and PFE profiles with and without a 3.5y mandatory break clause.

The impact of the **Mandatory Break Clause** is observed as a deterministic drop in both profiles at the 3.5-year mark (late 2011). Up to this date, the exposure follows the base case; however, upon the mandatory termination, the trade is settled at its prevailing Mark-to-Market value, and the exposure is reduced to zero for the remainder of the 10-year horizon. This highlights the effectiveness of mandatory breaks in shortening the legal maturity and providing certain capital relief.

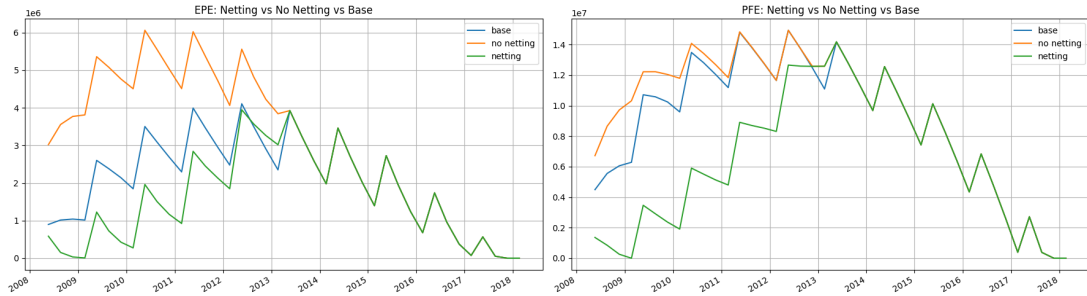


Figure 4: EPE and PFE profiles with and without netting agreement.

The **Netting Agreement** demonstrates a significant reduction in counterparty risk by offsetting the 10-year payer swap with a 5-year receiver swap.

- **Benefit:** In the first 5 years, the "netting" line is considerably lower than the "no netting" (gross) profile, as the two swaps partially hedge each other.
- **Convergence:** After the 5-year mark, the receiver swap expires, and the exposure converges back to the standalone "base" profile of the 10-year payer swap.

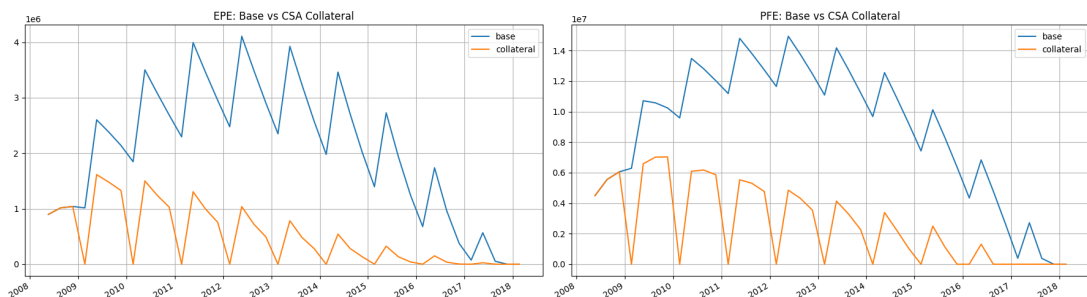


Figure 5: EPE and PFE profiles with and without CSA collateral.

The implementation of a CSA with annual margin calls results in a characteristic "sawtooth" pattern. Since the counterparty fully covers its exposure ($THR_C = 0$), the exposure is reset to zero at each annual collateral posting date. The residual exposure observed between the calls represents the risk arising from market movements during the annual interval, confirming that while collateral significantly mitigates risk, it does not eliminate it entirely due to the frequency of the margin calls.

6 CVA Estimation and Sensitivity Analysis

6.1 CVA Formula and Discretization

The Credit Valuation Adjustment (CVA) represents the expected discounted loss due to the possible default of the counterparty over the life of the transaction. Under a set of simplifying assumptions — constant Loss Given Default, independence between exposures and default events, and unilateral default (only counterparty C can default) — the CVA at time t_0 is given by:

$$\text{CVA}_0 = \text{LGD}_C \int_0^T \text{EPE}_0(t) B_0(t) S_C(0, t) dPD_C(t), \quad (11)$$

where $B_0(t)$ is the risk-free discount factor, $S_C(0, t)$ is the survival probability of counterparty C up to time t , and $dPD_C(t)$ is the incremental default probability of C at time t , conditional on C having survived up to t .

Flat hazard rate model. We model the default time τ of C as an exponential random variable with constant intensity λ :

$$\tau \sim \text{Exp}(\lambda), \quad S_C(0, t) = \text{Prob}(\tau > t) = e^{-\lambda t}.$$

The conditional default probability in $[t, t + dt]$, given that C survived up to t , is

$$dPD_C(t) = \lambda dt.$$

This follows directly from the memoryless property of the exponential distribution: the instantaneous probability of default per unit of time is constant and equal to λ , regardless of the time already elapsed. Therefore, the product $S_C(0, t) \cdot dPD_C(t)$ represents the unconditional marginal probability of default in $[t, t + dt]$:

$$S_C(0, t) \cdot dPD_C(t) = e^{-\lambda t} \cdot \lambda dt = \lambda e^{-\lambda t} dt,$$

and the CVA formula (11) simplifies to

$$\text{CVA}_0 = \text{LGD}_C \int_0^T \text{EPE}_0(t) B_0(t) \lambda e^{-\lambda t} dt. \quad (12)$$

Discretization. The integral in (12) is evaluated numerically on the 40 equally spaced simulation dates $0 = t_0 < t_1 < \dots < t_{40} = T$. Partitioning the interval $[0, T]$ into sub-intervals $[t_{i-1}, t_i]$ and assuming $\text{EPE}_0(t)$ and $B_0(t)$ to be approximately constant on each sub-interval at the value taken at the right endpoint t_i , the integral over $[t_{i-1}, t_i]$ becomes:

$$\int_{t_{i-1}}^{t_i} \text{EPE}_0(t) B_0(t) \lambda e^{-\lambda t} dt \approx \text{EPE}_0(t_i) B_0(t_i) \int_{t_{i-1}}^{t_i} \lambda e^{-\lambda t} dt.$$

The remaining integral has a closed-form solution:

$$\int_{t_{i-1}}^{t_i} \lambda e^{-\lambda t} dt = [-e^{-\lambda t}]_{t_{i-1}}^{t_i} = e^{-\lambda t_{i-1}} - e^{-\lambda t_i} =: \Delta PD_i,$$

which is precisely the unconditional probability of default in $[t_{i-1}, t_i]$. Summing over all sub-intervals yields the discrete CVA approximation:

$$\boxed{\text{CVA}_0 \approx \text{LGD}_C \sum_{i=1}^{40} \text{EPE}_0(t_i) B_0(t_i) (e^{-\lambda t_{i-1}} - e^{-\lambda t_i})}. \quad (13)$$

Implementation details. The EPE profile $\{\text{EPE}_0(t_i)\}$ is taken directly from the Monte Carlo output described in Section 4. The risk-free discount factors $\{B_0(t_i)\}$ are obtained from the bootstrapped initial curve via linear interpolation of zero rates. The time fractions t_i are computed using the ACT/365 convention (physical time), consistently with the dynamics of the Hull–White state variable. The parameters used are $\text{LGD}_C = 0.60$ and $\lambda = 200$ bps.

6.2 CVA Results

Table 1 reports the CVA computed under each of the five scenarios analyzed in Section 5.

Scenario	CVA (€)	Change vs Base
Base	190,293	—
Break Clause	84,182	−55.8%
No Netting	304,593	+60.1%
Netting	141,460	−25.7%
Collateral (CSA)	49,822	−73.8%

Table 1: CVA estimates across the five scenarios. Parameters: LGD = 60%, $\lambda = 200$ bps

The results confirm the expected hierarchy of risk mitigants. The **base case** CVA of €190,293 represents the unmitigated cost of counterparty default risk embedded in the 10-year payer swap.

The **mandatory break clause** reduces the CVA by approximately 56%, reflecting the deterministic elimination of all exposure beyond the 3.5-year termination date. Since no exposure can accumulate after the break, the integral in (13) effectively truncates at $t = 3.5$ y, cutting off the most significant portion of the EPE profile.

The **netting agreement** reduces the CVA by approximately 26% relative to the base case. The benefit arises from the sub-additivity of the positive-exposure operator: the net MtM of the two swaps is smaller than the sum of their individual positive exposures, particularly during the first five years when both contracts are alive and partially offset each other. After the 5-year receiver swap expires, the netting benefit disappears and the CVA converges toward the base case contribution from the remaining payer swap.

The **no-netting** scenario, by contrast, computes the CVA as the sum of the individual positive exposures of the two swaps, yielding a value 60% above the base case. This confirms the sub-additivity property: $\text{CVA}^{\text{netting}} \leq \text{CVA}^{\text{base}} \leq \text{CVA}^{\text{no netting}}$.

The **CSA with annual margin calls** achieves the largest reduction, cutting the CVA by approximately 74%. Despite the relatively low margin call frequency (annual rather than daily), the combination of a zero threshold for the counterparty and full collateralisation at each posting date resets the exposure to near zero at each anniversary. The residual CVA of €49,822 reflects the exposure accumulated between consecutive margin calls, during which the MtM can move freely without triggering a collateral adjustment.

6.3 CVA Sensitivity Analysis

The CVA computed in (13) depends on three categories of market and model parameters: the *credit parameters* (λ , LGD), the *interest-rate curve* ($B_0(t)$ and, through the EPE, the simulated discount factors), and the *Hull–White volatility* (σ). To quantify the sensitivity of the base-case CVA to each of these risk factors, we apply a standard **bump-and-reprice** approach: each parameter is shocked by a small amount $\Delta\theta$ and the CVA is recomputed, keeping all other inputs fixed:

$$\frac{\partial \text{CVA}}{\partial \theta} \approx \frac{\text{CVA}(\theta + \Delta\theta) - \text{CVA}(\theta)}{\Delta\theta}.$$

The five sensitivities computed are described below.

- **CS01 — sensitivity to λ (+10 bps):** the hazard rate is increased by 10 bps. This does not require re-running the Monte Carlo simulation, since λ only enters the discretized CVA formula through the default probabilities $\Delta PD_i = e^{-\lambda t_{i-1}} - e^{-\lambda t_i}$.
- **LGD sensitivity (+10%):** the LGD is increased from 60% to 66%. Since LGD enters the CVA formula linearly as a multiplicative factor, the sensitivity equals exactly $\text{CVA}_{\text{base}} \times \Delta \text{LGD} / \text{LGD}$, providing a useful cross-check.
- **IR Delta 0–5y (+1 bp):** the zero rates of the bootstrapped curve are shifted upward by 1 bp for all maturities up to 5 years. The bumped discount factors are used both to recompute the EPE (via a new affine-trick evaluation) and to re-discount the CVA integral.
- **IR Delta 5–10y (+1 bp):** same as above, applied to the 5–10 year portion of the curve.
- **Vega — sensitivity to σ (+1 bp):** the Hull–White volatility parameter is increased by 1 bp. This requires re-simulating the OU paths and recomputing the full EPE profile before recalculating the CVA.

Table 2 and Figure 6 summarise the results.

Risk Factor	Shock	ΔCVA (€)	Sign
Lambda (λ)	+10 bps	+8.717,01	Positive
LGD	+10%	+19.029,26	Positive
IR Delta 0–5y	+1 bp	–712,60	Negative
IR Delta 5–10y	+1 bp	+2.323,69	Positive
Vega (σ)	+1 bp	+1.977,43	Positive

Table 2: CVA sensitivities — bump-and-reprice on the base case CVA of €190,293.

Credit sensitivities. Both λ and LGD carry a positive sensitivity, as expected: a higher probability of default or a larger loss given default directly increases the expected loss embedded in the CVA. The LGD sensitivity of €19,029 is consistent with the theoretical prediction: a 10% relative increase in LGD (from 60% to 66%) should produce exactly a 10% increase in CVA, i.e. $190,293 \times 0.10 = 19,029$ €. This serves as a numerical sanity check on the implementation.

Interest-rate sensitivities. The two IR deltas exhibit opposite signs, reflecting the different role of discounting and exposure along the maturity spectrum.

For the **0–5y bucket**, a +1 bp shock produces a *negative* ΔCVA of €–712. In the short end of the curve, the dominant effect is the reduction of the risk-free discount factors $B_0(t_i)$ entering the CVA integral: discounting the near-term EPE more heavily reduces the CVA. The offsetting effect — a payer swap going slightly more in-the-money as rates rise, which increases EPE — is present but smaller in the short end, where the EPE profile is still low and rising.

For the **5–10y bucket**, the same shock produces a *positive* ΔCVA of €+2,323. In this part of the curve the EPE is near its peak. A rate increase makes the payer swap more in-the-money, increasing the EPE significantly. This effect dominates the discounting effect, producing a net positive sensitivity.

This asymmetry reflects the fundamental structure of a payer IRS under diffusive interest-rate dynamics: the EPE profile peaks in the 3–4 year region and is most sensitive to rate movements in the medium-to-long part of the curve, while the short end primarily affects discounting.

Vega. The positive sensitivity of €+1,977 per basis point of σ is consistent with the theoretical expectation. A higher Hull–White volatility widens the distribution of simulated x_t paths, generating more scenarios with large positive MtM values. Since the EPE takes the mean of positive exposures, it increases with volatility due to the convexity of the $\max(\cdot, 0)$ operator: larger dispersion always raises the expected value of the positive part, regardless of the direction of the shift.

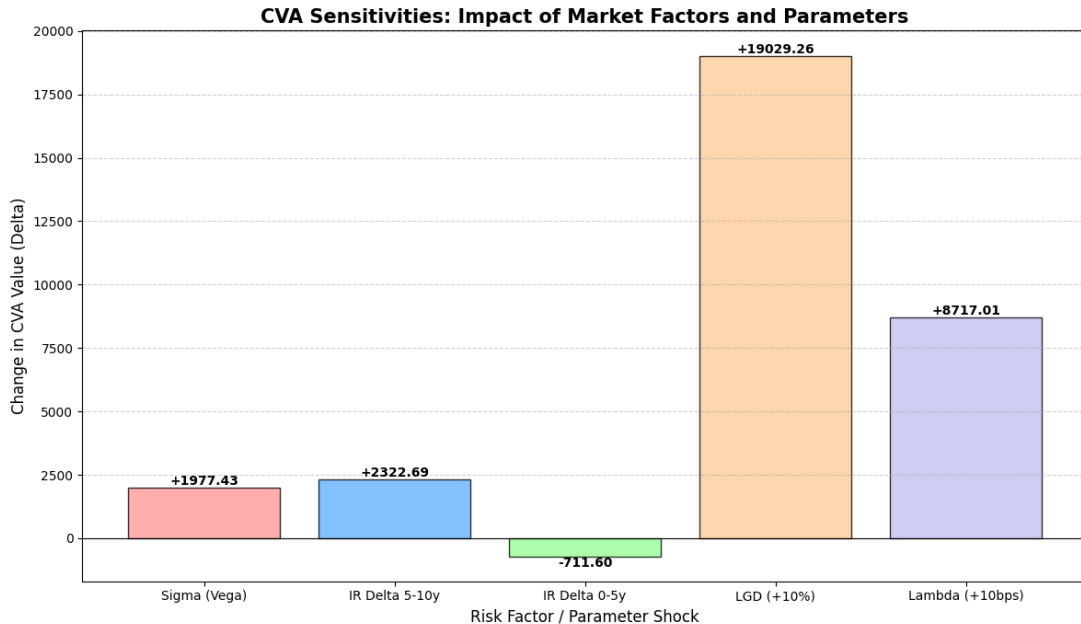


Figure 6: CVA sensitivities: impact of a one-unit shock to each risk factor on the base-case CVA.

Appendix: Errors Found in the Skeleton Code

1. `simulations_num = 100_000` instead of 250,000.
2. `simulations_expiry` computed with `month_offset=10` instead of `year_offset=10`.

3. ACT/360 instead of ACT/365 in `simulate_ou_process`.
4. Fixed-leg cash flows not multiplied by δ_i ; critical for the quarterly swap where $\delta_i \approx 0.25$.